CHAPTER 13

Monopoly

ECON101 - Prince Sultan University

features common across market structure

- sellers
- Product
- Entry
- control over price

→ extreme of perfect

OVERVIEW

Learning objectives

Some markets have only one seller. This chapter looks at how a firm comes to be the only seller of a product, and at how such a firm decides how much to produce and what price to charge.

1. Explain how monopoly arises. → no close substitutes + barriers of entry
2. Explain how a single-price monopoly chooses its output and price.
3. Explain why a monopoly's price is higher than its marginal cost, and why its economic profit can last. → YES in long run bc of high barrier entry so new rivals are unable to enter the market

Running example: Sahra Tours, a fictional firm with the only right to run guided tours of a desert canyon.

What is a monopoly?

A **monopoly** is a market with a single firm that produces a good or service with no close substitutes and that is protected by a barrier that prevents other firms from entering that market.

So every monopoly shows two features:

1. No close substitutes

Buyers have no good alternative product to turn to, so the firm does not compete with the sellers of other products.

2. A barrier to entry

Something keeps new firms out, so no rival can start selling the same product.

Compare perfect competition (Chapter 12): many firms, identical products and free entry.

No close substitutes, and barriers to entry

No close substitutes

A product has a close substitute when buyers can switch to another product that meets the same need almost as well. If one exists, a single seller still faces competition from the firms that make the substitute.

Example: for lighting, appliances and air conditioning, nothing practical can replace the electricity supplied to a home, so it has no close substitute for these uses.

Barriers to entry

A **barrier to entry** is any constraint that protects a firm from potential competitors.

Barriers come in three types:

- **Natural:** one firm can supply the market more cheaply than several firms.
- **Ownership:** one firm controls most of a key resource. e.g. diamonds in mines
- **Legal:** the government restricts entry through a public franchise, a government license, a patent or a copyright.

rights / restrictions offered by the government

e.g. patents rights claimed by authorized firms
if anyone else attempts to claim it they will face legal costs

MONOPOLY AND HOW IT ARISES

Natural barriers: natural monopoly

A **natural monopoly** is a market in which one firm can supply the entire market at a lower cost than two or more firms could, because its cost per unit keeps falling as its output grows.

Examples: the networks of pipes and cables that deliver water, gas and electricity to the homes in a city.

A huge cost to build

Laying pipes or cables under every street costs a great deal.

A small cost to serve more

Once the network exists, one more home adds little cost, so cost per unit falls.

One network is cheapest

A second network would double the building cost but serve the same homes.

A second firm could not match the first firm's low cost per unit, so the market has room for only one firm.

Ownership and legal barriers

Ownership barrier: one firm owns most of a key resource, such as almost all the known deposits of a rare mineral, so rivals cannot get enough of it to compete.

Legal barrier: the government restricts competition and entry, creating a **legal monopoly**, through one of four instruments:

Public franchise: one firm's exclusive right to supply a good or service, such as a city's bus routes.

Government license: permission needed to enter some professions; pharmacists and doctors must be licensed.

Patent: an inventor's exclusive right to a new product or production method, such as a new medicine.

Copyright: a creator's exclusive right to a book, a song, a film, an app or another creative work.

Patents encourage invention and innovation: inventors can earn a return from putting their new ideas to use.

Running example: Sahra Tours

Sahra Tours (a fictional firm) holds a **public franchise**: the government has given it the only right to run guided tours of a protected desert canyon. No other firm may offer these tours, and there is no other practical way to see the canyon. Its output is measured in tours per day, and all amounts are in SAR.

A monopoly

No close substitute: visitors cannot see the canyon any other way.

Barrier to entry: a legal one, a public franchise.

A single-price monopoly

A **single-price monopoly** is a firm that must sell each unit of its output for the same price to all its customers.

Every tour on a given day sells at the same price.

A price setter

As the only seller, it faces the **market demand** for tours. To sell more tours a day, it must lower its price.

So it is a **price setter**, whereas Al-Bustan Farm in Chapter 12 was a price taker.

A SINGLE-PRICE MONOPOLY'S OUTPUT AND PRICE

Price and marginal revenue

→ in Perfect competition MR is same since price doesn't change

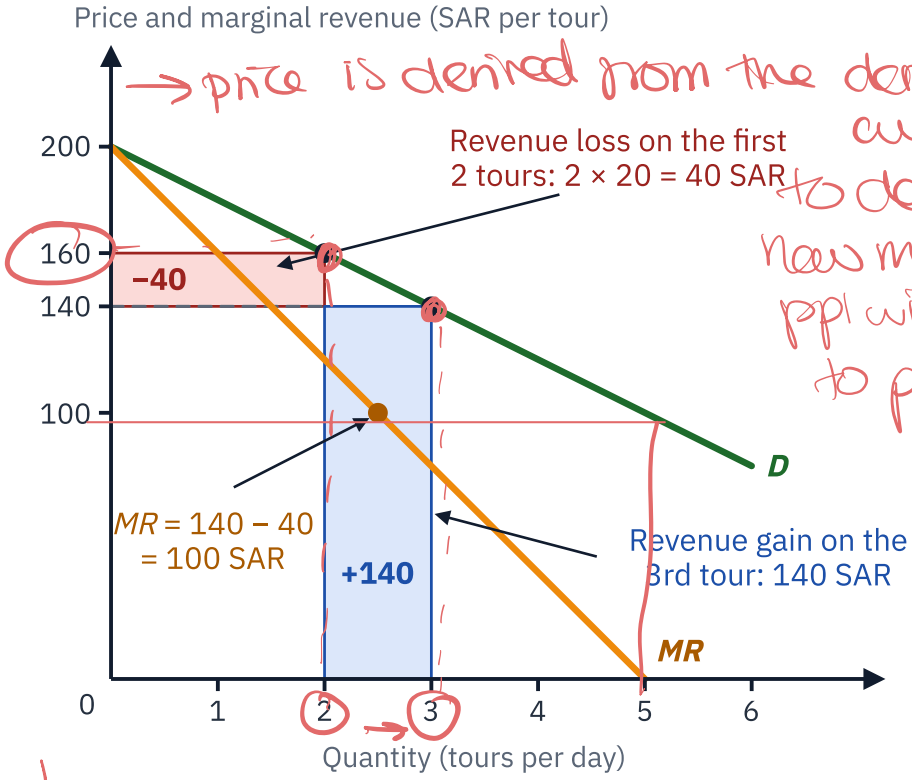
Monopoly is sometimes inefficient assuming its TR increases

Quantity (*Q*, tours per day)	Price (*P*, SAR per tour)	Total revenue (*TR*, SAR)	Marginal revenue (*MR*, SAR per extra tour)
0	200	0	
			180
1	180	180	
			140
2	160	320	
			100
3	140	420	
			60
4	120	480	
			20
5	100	500	
			−20
6	80	480	

MR < P

- The table is the market demand schedule: the lower the price, the more tours people buy each day.
- **Total revenue:** $TR = P \times Q$. At 160 SAR a tour, 2 tours bring in 320 SAR.
- **Marginal revenue** is the change in total revenue from selling one more unit: $MR = \Delta TR / \Delta Q$.
- Each *MR* value sits between the two rows it links. From 2 to 3 tours, *TR* rises from 320 to 420 SAR, so *MR* = 100 SAR.
- From the 2nd tour on, *MR* is **less than the price**: the 3rd tour sells for 140 SAR but adds only 100 SAR to revenue.

The demand and marginal revenue curves



- *D* is the market demand curve. The *MR* curve is plotted from the table, each value halfway between the two outputs it links.
- Lowering the price from 160 to 140 SAR raises sales from 2 to 3 tours.
- **Red area:** the first 2 tours now sell for 20 SAR less, a loss of 40 SAR.
- **Blue bar:** the 3rd tour brings in 140 SAR.
- So *MR* = 140 − 40 = 100 SAR. At every positive output, the *MR* curve lies **below** the demand curve.

A SINGLE-PRICE MONOPOLY'S OUTPUT AND PRICE

Why marginal revenue is less than price

To sell one more tour, Sahra Tours must lower the price of **every** tour, not only the extra one. Two opposing forces then act on total revenue:

Revenue gain (+)

The extra tour is sold at the new, lower price.

From 2 to 3 tours: the 3rd tour adds **140 SAR**.

Revenue loss (−)

The tours that were already being sold now bring in less each.

From 2 to 3 tours: the first 2 tours each sell for 20 SAR less, so revenue falls by **2 × 20 = 40 SAR**.

***MR* = gain − loss = 140 − 40 = 100 SAR, less than the price of 140 SAR.**

- Beyond the first tour there is always such a loss, so the *MR* curve lies below the demand curve. From 5 to 6 tours the loss, 5 × 20 = 100 SAR, exceeds the gain of 80 SAR, so *MR* = −20 SAR.
- A perfectly competitive firm has no revenue loss: it can sell more at the same market price, so its *MR* equals the price.

A SINGLE-PRICE MONOPOLY'S OUTPUT AND PRICE

Choosing output and price

A monopoly's costs behave just like those of a competitive firm; what differs is the demand it faces. It looks for the output and price that make economic profit, *TR* – *TC*, as large as possible, in two steps:

Step 1: choose the quantity

Produce the output at which ***MR* = *MC***.

While *MR* exceeds *MC*, one more tour adds more to revenue than to cost, so profit rises. Once *MC* exceeds *MR*, one more tour lowers profit.

Step 2: set the price

Charge the **highest price buyers will pay** for that quantity.

Read it from the demand curve, directly above the chosen output. The firm does not charge the highest price anyone will pay: at 180 SAR it would sell only 1 tour.

Every firm that produces chooses the output where *MR* = *MC*. The difference: for a perfectly competitive firm *P* = *MR*, but for a monopoly *P* > *MR*.

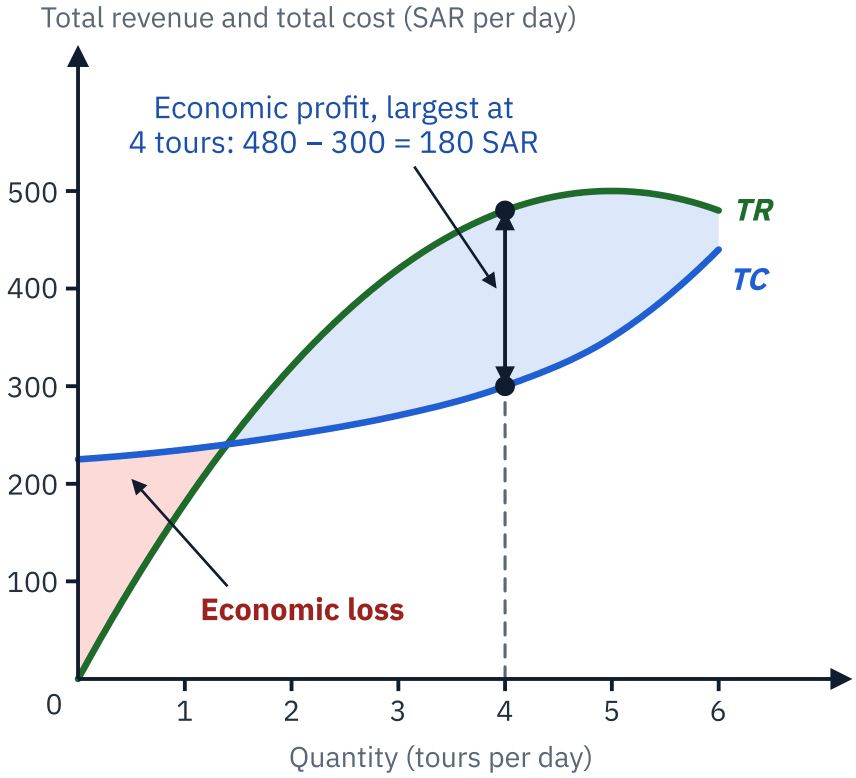
Sahra Tours: revenue, cost and profit

TR, *TC* and economic profit in SAR per day; price, *ATC*, *MR* and *MC* in SAR per tour

Quantity (tours)	Price (*P*)	*TR*	*MR*	*TC*	*MC*	*ATC*	Economic profit
0	200	0		225			−225
			180		10		
1	180	180		235		235	−55
			140		15		
2	160	320		250		125	70
			100		20		
3	140	420		270		90	150
			60		30		
4	**120**	**480**		**300**		**75**	**180**
			20		50		
5	100	500		350		70	150
			−20		90		
6	80	480		440		73.3	40

- Total fixed cost is 225 SAR a day (vehicles and the franchise fee).
- From 3 to 4 tours, *MR* = 60 > *MC* = 30: profit rises by 30 SAR, from 150 to 180.
- From 4 to 5 tours, *MR* = 20 < *MC* = 50: profit falls by 30 SAR, from 180 to 150.
- So profit is largest at **4 tours**, sold at **120 SAR** each: 480 − 300 = **180 SAR** a day.

Total revenue and total cost curves

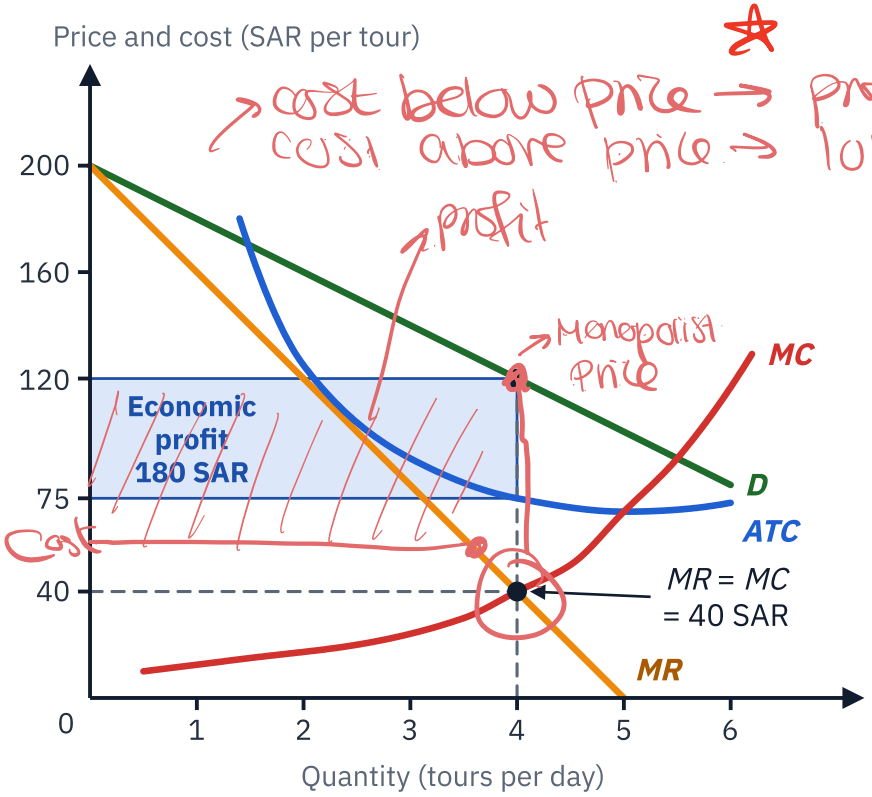


- *TR* rises by less and less with each extra tour, because the price must fall to sell more. It peaks at 500 SAR with 5 tours and then falls.
- *TC* starts at the fixed cost of 225 SAR and rises more and more steeply, because each extra tour costs more than the one before.
- At 0 and 1 tour, revenue does not cover total cost (at 1 tour, *TR* = 180 < *TC* = 235): an economic loss.
- Economic profit, the gap *TR* – *TC*, is widest at **4 tours**: 480 – 300 = 180 SAR.

need to know where MR-MC intersect

A SINGLE-PRICE MONOPOLY'S OUTPUT AND PRICE

Output, price and economic profit



- **Output:** *MR* = *MC* at **4 tours**; on the smooth curves, both are at 40 SAR there.
- **Price:** go up to the demand curve. Buyers will pay **120 SAR** a tour for 4 tours a day.
- **Cost:** *ATC* at 4 tours is 75 SAR, so each tour earns 120 – 75 = 45 SAR more than its average cost.
- **Blue rectangle:** economic profit = (*P* – *ATC*) × *Q* = 45 × 4 = **180 SAR** a day, the same maximum as in the table.

MR = MC → Quantity
Price → Demand Curve

Profit = (P - ATC) × Q
(120 - 75) × 4

A SINGLE-PRICE MONOPOLY'S OUTPUT AND PRICE

A monopoly's price exceeds marginal cost

Every firm that produces chooses the output where *MR* = *MC*. What differs is how price compares with marginal revenue:

Perfect competition (Chapter 12)

The firm sells any quantity at the market price, so *P* = *MR*.

P* = *MR* = *MC

Al-Bustan Farm: 60 = 60 = 60 SAR at 7 boxes.

Single-price monopoly

To sell more, the firm must lower its price, so *P* > *MR*.

P* > *MR* = *MC

Sahra Tours: 120 > 40 = 40 SAR at 4 tours.

On Sahra Tours' smooth curves, *MR* and *MC* both stand at 40 SAR at 4 tours, while buyers pay 120 SAR a tour. Price is above *MR* because of the revenue loss on the tours already being sold.

Can a monopoly's profit last?

Barriers block entry

In perfect competition, an economic profit attracts new firms. In monopoly, a barrier to entry keeps them out, so the economic profit **can continue, even in the long run**. No other firm may run tours of the canyon.

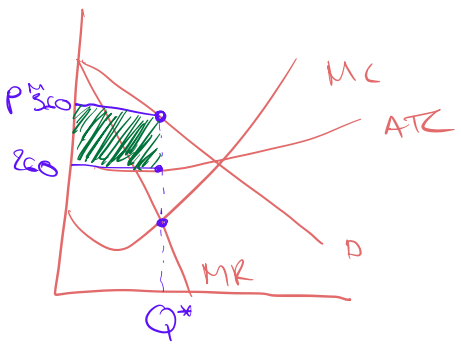
Higher fixed cost

Suppose the franchise fee rises, raising fixed cost by 180 SAR a day. *MR* and *MC* do not change, so output and price stay the same and only profit falls, to zero.

With a larger rise, the firm makes an economic loss. If that loss is permanent, it leaves the market.

Sahra Tours at its best output, SAR per day

	Before	After
Total fixed cost	225	405
Output (tours)	4	4
Price (SAR per tour)	120	120
TR	480	480
TC	300	480
Economic profit	**180**	**0**



5 (where MR = MC)

If no calculation needed, simply label curves and coordinates

Profit = (P − ATC) × Q

= (300 − 200) × 5

= 100 × 5 = 500

SUMMARY

Key takeaways

- A **monopoly** is a market with one firm, selling a product with no close substitutes and protected by a **barrier to entry**: natural, ownership or legal.
- A single-price monopoly faces the market demand, so to sell more it must lower its price. Its **marginal revenue is less than its price**.
- It produces where ***MR = MC*** and charges the highest price buyers will pay for that output: 4 tours at 120 SAR for Sahra Tours.
- So its price exceeds marginal cost: ***P > MR = MC***.
- Barriers to entry keep rivals out, so the monopoly's economic profit can last. A rise in fixed cost lowers profit but leaves output and price unchanged.

SOURCE

Source

Michael Parkin (2022) *Microeconomics, Global Edition*, 14th Edition, published by Pearson Education.